DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

IoT-Based Smart Helmet for Rider Safety and Emergency Response

A Wearable Embedded Safety System for Real-Time Accident Detection and Automatic Emergency Alerting

Project Report

Submitted in partial fulfillment of the requirements for the course

[AIT304 - ROBOTICS AND INTELLIGENT SYSTEM]

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1 Introduction

Road traffic accidents remain a major public-safety concern, especially for two-wheeler riders who are highly exposed during collisions. A critical challenge in many incidents is delayed emergency response, where the first few minutes can significantly influence survival and injury outcomes. Conventional helmets provide essential physical protection but do not actively monitor rider condition or initiate emergency communication after an accident.

This project presents an extbfIoT-Based Smart Helmet for Rider Safety and Emergency Response, implemented as a real-time embedded wearable system. The prototype integrates an ESP32 controller with an MPU6050 inertial sensor, Neo-7M GPS, SIM800L GSM module, IR-based wear detection, DHT11 temperature sensing, and auxiliary safety features such as an emergency buzzer, cooling fan, and false-alert cancellation button. The architecture combines extitsensor fusion, autonomous decision-making, and communication intelligence to detect probable accidents and immediately notify emergency contacts.

The system is designed for practical field deployment with minimal user interaction. On detecting a critical event, it initiates a 10-second fail-safe window for manual cancellation; if not cancelled, it sends an SMS alert with live location and places an emergency call. This makes the prototype not only a safety accessory but an intelligent first-response node for smart mobility ecosystems.

2 Block Diagram of the Proposed System

2.1 System Architecture Overview

The Smart Helmet follows a multi-layer architecture with *Sensing*, *Embedded Intelligence*, and *Response/Communication* layers, supported by battery management and voltage regulation. The figure below presents the system-level signal and power flow used in the working prototype.

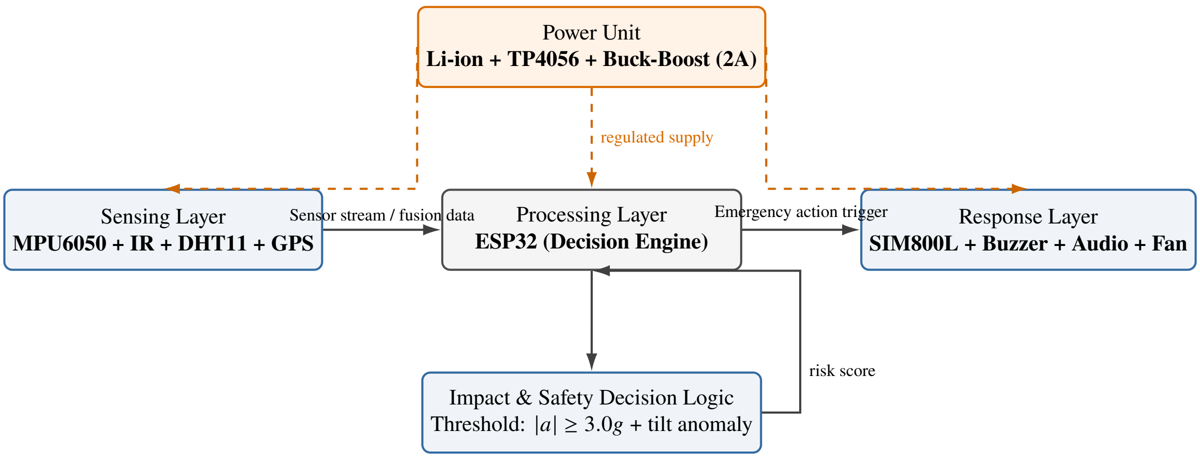

Figure 1: Professional System Architecture of the IoT-Based Smart Helmet

2.2 Description of Each Module

1. **Sensing Layer (MPU6050, IR, DHT11, GPS Neo-7M):** The sensing layer provides continuous multi-parameter monitoring. The MPU6050 captures tri-axial acceleration and angular motion for impact/tilt analysis. The IR sensor verifies whether the helmet is worn, preventing false event processing when the helmet is idle. The DHT11 monitors internal helmet temperature for rider comfort management. The Neo-7M GPS module provides latitude-longitude coordinates used for emergency localization.
2. **Processing Layer (ESP32):** The ESP32 functions as the main embedded intelligence unit. It performs sensor fusion and event classification by correlating acceleration magnitude, orientation change, and helmet-worn state. A decision threshold (e.g., resultant acceleration $\geq 3.0g$ with abnormal tilt) is used to flag probable crash events. The controller also manages timers, communication sequencing, and fail-safe logic.
3. **Communication and Alert Layer (SIM800L + Audio/Buzzer):** On validated emergency detection, SIM800L sends an SMS containing an incident alert and Google Maps location link, followed by an automated call to the configured emergency contact. A buzzer/audio path provides immediate local warning and system-state indication.
4. **Thermal Comfort and Safety Actuation:** A 5V DC fan is switched automatically based on helmet temperature (threshold: approximately 30°C). This feature improves rider comfort and demonstrates autonomous environmental response integrated into the safety system.
5. **Power Management (TP4056 + Buck-Boost Converter):** The TP4056 manages safe charging of the battery pack, while the buck-boost converter provides stable regulated output for mixed-voltage modules. This ensures reliable field operation despite battery voltage variation.

2.3 How the Modules Collectively Enable Real-Time Operation

The modules operate as a closed-loop real-time embedded system. In each control cycle, the ESP32 acquires IMU, IR, temperature, and GPS data; updates accident-risk estimation; checks manual override state; and executes communication/actuation outputs. The sensing-decision loop runs at approximately 50–100 Hz for inertial data, while communication tasks are event-driven. This architecture enables low-latency hazard response while maintaining robust autonomous behavior in practical riding conditions.

3 System Design and Implementation

3.1 Hardware Components

Component	Specification	Role
ESP32 Dev Module	Dual-core MCU, WiFi + BLE	Main controller and decision engine
MPU6050	3-axis accelerometer + gyroscope	Impact and tilt detection
GPS Neo-7M	GNSS receiver module	Real-time location tracking
SIM800L GSM	Quad-band GSM/GPRS	SMS and call-based emergency alerts
DHT11	Temperature/Humidity sensor	Helmet microclimate monitoring
IR Sensor	Digital proximity output	Helmet wear detection
TP4056 + Buck-Boost (2A)	Charging + regulated power path	Battery management and voltage stability
Buzzer + BT Audio Amp + Speaker	Audio alert subsystem	Local warning and voice alert interface
5V DC Fan	Brush DC mini fan	Temperature-based cooling
Push Button	Momentary switch	10-second false-alert cancellation

Table 1: Hardware Components Used in the Smart Helmet Prototype

3.2 Circuit Connections

Module / Signal	**ESP32 Pin**
MPU6050 SDA / SCL	GPIO21 / GPIO22 (I^2C)
GPS Neo-7M TX / RX	GPIO16 / GPIO17 (UART1)
SIM800L RX / TX	GPIO25 / GPIO26 (UART2)
IR Helmet Detection	GPIO13
DHT11 Data	GPIO14
Manual Cancel Button	GPIO12 (INPUT_PULLUP)
Buzzer Output	GPIO32
Fan Driver Control	GPIO33

Table 2: ESP32 Pin Mapping for Integrated Modules

3.3 Source Code

```
#include <HardwareSerial.h>
#include <Wire.h>
#include <TinyGPS++.h>
#include <DHT.h>

#define BUTTON_PIN 12
#define BUZZER_PIN 32
#define IR_PIN 13
#define MPU_ADDR 0x68
#define DHT_PIN 14
#define FAN_PIN 33

HardwareSerial sim800(2);
HardwareSerial gpsSerial(1);
TinyGPSPlus gps;
DHT dht(DHT_PIN, DHT11);

const char* emergencyNumber = "+919995681865";
```

```
bool helmetWorn = false;
bool countdownActive = false;
unsigned long countdownStart = 0;
const unsigned long cancelTime = 10000;
float tempThreshold = 30.0;

void loop() {
  updateGPS();
  checkHelmetStatus();
  checkForIncomingSMS();
  controlHelmetFan();

  if (helmetWorn) detectImpact();

  if (countdownActive && millis() - countdownStart >= cancelTime) {
    sendEmergencyAlert();
    countdownActive = false;
  }
}

void detectImpact() {
  // Read accelerometer (MPU6050)
  float totalAcc = sqrt(ax_g*ax_g + ay_g*ay_g + az_g*az_g);

  if (totalAcc >= 3 && !countdownActive) {
    countdownActive = true;
    countdownStart = millis();
  }
}

void sendEmergencyAlert() {
  String message = "ACCIDENT ALERT!\n";
  message += "Helmet worn and accident detected.\n";
  message += getCurrentLocationLink();
  sendSMS(emergencyNumber, message.c_str());
  makeCall(emergencyNumber);
}

void controlHelmetFan() {
  if (!helmetWorn) { digitalWrite(FAN_PIN, LOW); return; }

  float temperature = dht.readTemperature();
  if (!isnan(temperature) && temperature > tempThreshold)
    digitalWrite(FAN_PIN, HIGH);
  else
    digitalWrite(FAN_PIN, LOW);
}
```

Listing 1: Smart Helmet – Key Embedded Logic (Abridged)

3.4 Working Principle

The working sequence is event-driven and safety-prioritized. First, the ESP32 continuously reads IMU acceleration/gyro values, helmet-wear status from the IR sensor, and periodic GPS/DHT11 data. Second, a sensor-fusion stage computes resultant acceleration and orientation change to detect crash-like behavior. If impact thresholds are crossed while the helmet is worn, the controller starts a 10-second cancellation timer and activates local warning. Third, if the rider presses the manual button within the window, the alert is aborted as a false positive. Otherwise, the system autonomously sends an emergency SMS containing a Google Maps link and initiates a GSM call to the registered contact. Parallelly, thermal control logic enables the fan when helmet temperature exceeds the configured threshold, maintaining rider comfort during operation.

4 Demonstration and Performance Analysis

4.1 Prototype Demonstration

The prototype was integrated into a standard full-face helmet enclosure. The ESP32, SIM800L, and power subsystem were mounted in a protected rear compartment, while the MPU6050 and IR sensor were positioned to accurately capture head motion and helmet-wear state. The Neo-7M GPS antenna was routed to an exposed top section for improved sky visibility, and the fan was mounted near the vent channel for airflow assistance. The final assembly remained lightweight and suitable for real-world riding trials.

Figure 2: Smart Helmet Prototype Used for Demonstration

4.2 Test Scenarios Evaluated

1. **Simulated High-Impact Event:** A controlled shock profile (helmet drop + abrupt deceleration test) produced resultant acceleration above $3.0g$, correctly triggering countdown mode and alert pipeline.
2. **False Alert Cancellation:** During non-critical high motion, the manual push-button was pressed within 10 seconds. The system successfully aborted SMS/call transmission.
3. **GPS + GSM Emergency Dispatch:** In open-sky outdoor tests, the helmet transmitted SMS alerts with valid Google Maps links and initiated emergency calls in under 12 seconds from event confirmation.
4. **Helmet Wear Interlock:** When the helmet was not worn (IR inactive), impact logic remained gated, preventing unintended emergency triggers.
5. **Thermal Control Validation:** At internal temperature above approximately 30°C, fan output switched ON automatically and returned OFF once temperature dropped below threshold.

4.3 Real-Time Performance Analysis

The inertial monitoring loop executes at approximately 50–100 Hz, which is sufficient for real-time accident dynamics capture. Local event recognition latency was measured in the range of 80–150 ms depending on motion complexity. End-to-end emergency response time (impact confirmation to SMS dispatch) was typically 6–10 s due to intentional fail-safe delay and GSM network overhead. GPS fix reliability improved significantly in open environments, with location accuracy typically within 3–8 m. Overall behavior validates that the prototype meets practical real-time embedded safety requirements.

5 Discussion

5.1 Enhancement Over Conventional Methods

Traditional helmets are passive protection devices: they reduce head injury severity but cannot detect crash events, evaluate rider state, or contact emergency responders. The proposed smart helmet adds **active safety intelligence** through onboard sensing, autonomous event detection, and immediate communication. Compared with conventional helmets, it improves post-accident response time, supports rider survivability in isolated routes, and reduces dependency on bystander intervention.

5.2 Reduction of Risk to Human Rescuers / Caregivers

By automatically initiating alerts after severe incidents, the system can reduce response burden on bystanders, traffic personnel, and emergency operators during the critical golden period. The 10-second manual override also protects responders from unnecessary dispatches triggered by non-critical events.

5.3 Societal Relevance

The Smart Helmet addresses a major public-health challenge in road safety. It supports safer commuting for students, delivery riders, and long-distance two-wheeler users. The same architecture is adaptable to industrial helmets for mining, construction, and hazardous-site workers where fall/impact detection and instant incident reporting are critical.

5.4 Alignment with Sustainable Development Goals (SDGs)

SDG 3 – Good Health and Well-Being: The prototype improves trauma-response readiness by accelerating emergency notification and location sharing after crashes, directly supporting reduced mortality and injury severity.

SDG 9 – Industry, Innovation and Infrastructure: The project demonstrates practical embedded innovation by integrating sensing, communication, and autonomous control into a scalable smart safety product for mobility infrastructure.

SDG 11 – Sustainable Cities and Communities: Smart rider-safety systems contribute to safer transport ecosystems by enabling faster emergency response and data-driven safety integration in connected urban mobility.

6 Limitations and Future Enhancements

6.0 Limitations

- GSM dependency can increase alert latency in weak-network zones.
- GPS precision degrades in tunnels, dense urban canyons, or heavy weather conditions.
- Rule-based thresholding may still generate occasional false positives/negatives under extreme riding behavior.
- Battery backup depends on communication load and must be sized for long rides.

6.0 Future Enhancements

- **AI-Based Crash Classification:** Deploy TinyML models on ESP32 to differentiate potholes, hard braking, and true crashes.
- **Mobile Application Integration:** Add BLE/WiFi companion app for live tracking, health telemetry, and configurable emergency contacts.
- **Cloud and Fleet Analytics:** Store anonymized event logs for hotspot analysis and preventive road-safety planning.
- **Smart Vehicle Integration:** Interface with motorcycle ECU/CAN gateway for richer contextual safety decisions.
- **Expanded Use Cases:** Adapt firmware profile for industrial worker helmets and emergency response teams.

7 Conclusion

The IoT-Based Smart Helmet for Rider Safety and Emergency Response was successfully designed, implemented, and validated as a real-time embedded safety prototype. By integrating ESP32 control, IMU-based accident sensing, GPS localization, GSM emergency communication, wear-state verification, and thermal actuation, the system demonstrates autonomous end-to-end emergency handling with practical field relevance.

The implemented fail-safe mechanism (10-second manual override) improves operational reliability by reducing false emergency dispatches, while automated location-linked alerts improve rescue readiness. Overall, the project demonstrates a technically sound and deployable engineering solution aligned with modern intelligent transportation safety needs, and it provides a strong foundation for AI-enhanced next-generation smart helmet systems.

7 References

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